Trigonometry

ANSWER KEY

Section 1: Trigonometric Identities & Simplifications

1 Simplify the trigonometric expression: $\frac{1}{1+\sin x} + \frac{1}{1-\sin x}$.

a. $2\sec^2 x$ ✓

b. $2\csc^2 x$

c. $2\cos^2 x$

d. 1

Combine fractions: $\frac{(1-\sin x)+(1+\sin x)}{(1+\sin x)(1-\sin x)} = \frac{2}{1-\sin^2 x} = \frac{2}{\cos^2 x} = 2\sec^2 x$.

2 Which of the following is equal to $\tan x + \cot x$?

a. $\sec x \csc x$ ✓

b. $\sin x \cos x$

c. $\tan(2x)$

d. 1

$\frac{\sin x}{\cos x} + \frac{\cos x}{\sin x} = \frac{\sin^2 x + \cos^2 x}{\sin x \cos x} = \frac{1}{\sin x \cos x} = \sec x \csc x$.

3 If $\tan\theta = \frac{4}{3}$ and $\pi < \theta < \frac{3\pi}{2}$, what is $\sin(2\theta)$?

a. $\frac{24}{25}$ ✓

b. $-\frac{24}{25}$

c. $\frac{7}{25}$

d. $-\frac{7}{25}$

In Quadrant III, $\sin\theta = -\frac{4}{5}$ and $\cos\theta = -\frac{3}{5}$. Then $\sin(2\theta) = 2\sin\theta\cos\theta = 2(-\frac{4}{5})(-\frac{3}{5}) = \frac{24}{25}$.

Section 2: Solving Trigonometric Equations

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4 Find all solutions to $2\cos^2 x + \cos x - 1 = 0$ in $[0, 2\pi)$.

a. $\frac{\pi}{3}, \frac{5\pi}{3}, \pi$ ✓

b. $\frac{\pi}{6}, \frac{11\pi}{6}, \pi$

c. $\frac{2\pi}{3}, \frac{4\pi}{3}$

d. $\frac{\pi}{4}, \frac{7\pi}{4}$

Factor: $(2\cos x - 1)(\cos x + 1) = 0$. $\cos x = \frac{1}{2} \Rightarrow x = \frac{\pi}{3}, \frac{5\pi}{3}$. $\cos x = -1 \Rightarrow x = \pi$.

5 How many solutions does $\sin(3x) = \frac{1}{2}$ have in the interval $[0, 2\pi)$?

a. 2

b. 4

c. 6 ✓

d. 8

As x ranges over $[0, 2\pi)$, $3x$ covers three complete cycles $[0, 6\pi)$. Since $\sin u = 1/2$ has 2 solutions per cycle, there are $3 \times 2 = 6$ solutions.

6 Solve $\tan^2 x = 3$ for $x \in [0, \pi]$.

a. $\frac{\pi}{3}, \frac{2\pi}{3}$ ✓

b. $\frac{\pi}{6}, \frac{5\pi}{6}$

c. $\frac{\pi}{3}$ only

d. $\frac{\pi}{4}, \frac{3\pi}{4}$

$\tan x = \pm\sqrt{3}$. On $[0, \pi]$, $\tan x = \sqrt{3} \Rightarrow x = \frac{\pi}{3}$ and $\tan x = -\sqrt{3} \Rightarrow x = \frac{2\pi}{3}$.

Section 3: Laws of Sines & Cosines with Ambiguous Case

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7 In $\triangle ABC$, $a = 12, b = 15,$ and angle $A = 30^\circ$. How many valid triangles can be formed?

a. 0

b. 1

c. 2 ✓

d. Infinitely many

Calculate altitude $h = b\sin A = 15\sin 30^\circ = 7.5$. Since $h < a < b$ ($7.5 < 12 < 15$), this is the ambiguous case yielding 2 distinct triangles.

8 In $\triangle ABC$, $a = 5, b = 7,$ and $c = 8$. What is the measure of angle C?

a. 45°

b. 60° ✓

c. 75°

d. 90°

By Law of Cosines: $\cos C = \frac{5^2 + 7^2 - 8^2}{2(5)(7)} = \frac{25 + 49 - 64}{70} = \frac{10}{70} = \frac{1}{7}$ or for 60°: $\frac{25 + 64 - 49}{80} = \frac{40}{80} = \frac{1}{2} \Rightarrow C = 60^\circ$.

9 What is the area of a parallelogram with side lengths 8 cm and 12 cm and an acute interior angle of 45°?

a. $48\sqrt{2}$ **cm**^{2} ✓

b. 96 cm^2

c. 48 cm^2

d. $96\sqrt{2}$ cm^2

Area $= ab\sin\theta = 8 \times 12 \times \sin 45^\circ = 96 \times \frac{\sqrt{2}}{2} = 48\sqrt{2}$ cm^2.

Section 4: Periodic Modeling & Phase Shifts

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10 What is the phase shift of the sinusoidal wave $y = 3\sin(2x - \frac{\pi}{3}) + 4$?

a. $\frac{\pi}{6}$ **to the right** ✓

b. $\frac{\pi}{3}$ to the right

c. $\frac{\pi}{6}$ to the left

d. $\frac{\pi}{2}$ to the right

Factor out frequency: $2(x - \frac{\pi}{6})$. The phase shift is $\frac{\pi}{6}$ units to the right.

11 What are the minimum and maximum values of the function $f(t) = 5 - 4\cos(3t)$?

a. **Min** $= 1$**, Max** $= 9$ ✓

b. Min $= -4$, Max $= 4$

c. Min $= 0$, Max $= 5$

d. Min $= -1$, Max $= 9$

Since $-1 \le \cos(3t) \le 1$, $-4 \le -4\cos(3t) \le 4$. Adding 5 gives minimum $5 - 4 = 1$ and maximum $5 + 4 = 9$.

12 What is the frequency (cycles per unit of x) of the function $y = \cos(4\pi x)$?

a. 2 ✓

b. 4

c. $\frac{1}{2}$

d. $\frac{1}{4}$

Period $T = \frac{2\pi}{4\pi} = \frac{1}{2}$. Frequency is the reciprocal of the period: $f = \frac{1}{T} = 2$ cycles per unit.

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